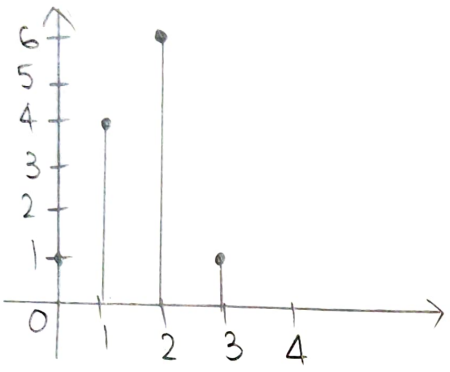
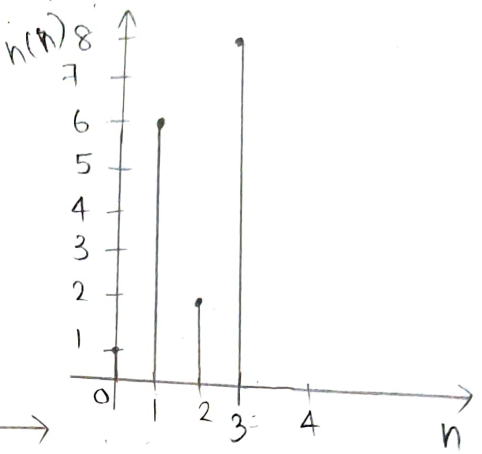
Linear convolution:-

$x = [1\ 4\ 6\ 1]$

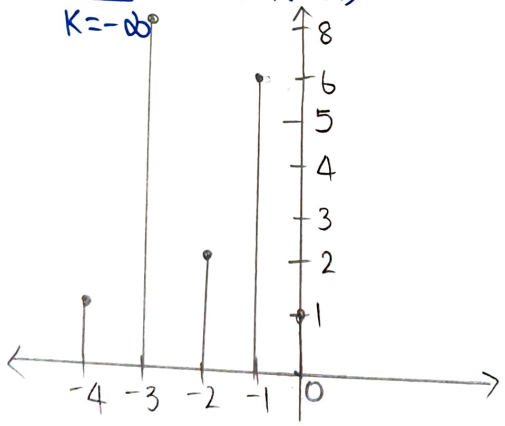
$h = [1\ 6\ 2\ 8]$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$$



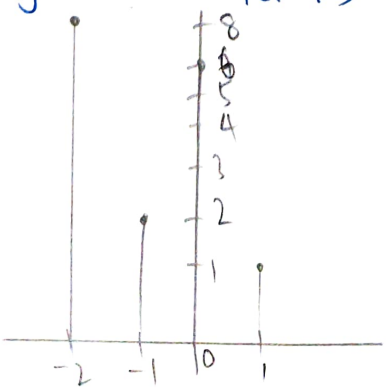


$$y(0) = \sum_{k=-\infty}^{\infty} x(k)\,h(-k)$$



$y(0) = (1 \times 1) = 1$

$y(1) = x(k)\,h(1-k)$



$= (1 \times 6) + (1 \times 4)$

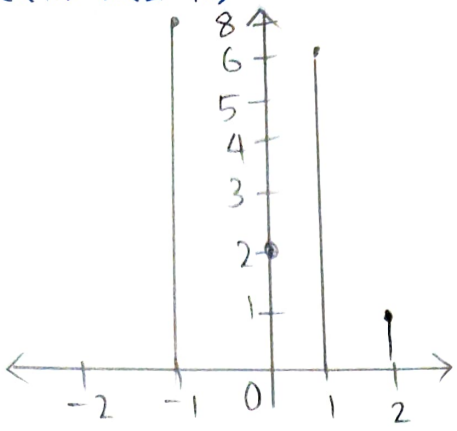
$= 6 + 4$

$y(1) = 10$

LINEAR CONVOLUTION PROGRAM:-

```
clc;
clear;
x=[1 4 6 1];
y=[1 6 3 8];
m=length(x);
n=length(h);
a=0:1:m-1;
b=0:1:n-1;
subplot(3,1,1);
plot2d3(a,x);
xlabel('Time');
ylabel('Amplitude');
title('Graphical Representation of input signal x');
subplot(3,1,2);
plot2d3(3,1,2);
xlabel('time');
ylabel('Amplitude');
title('Graphical Representation of Impulse signal h');
for i=1:n+m-1
conv:sum=0;
for j=1:i
if (((i+j+1)<=n)&(j<=m))
conv-sum=conv-sum+x(j)*h(i-j+1);
end;
y(i)=conv-sum;
end;
end;
```

$y(2) = x(k)\, h(2-k)$



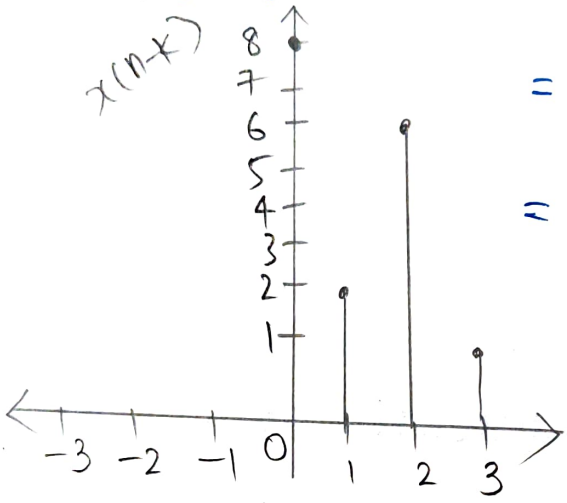
$= (1 \times 2) + (4 \times 6) + (1 \times 6)$

$= 2 + 24 + 6 + 8 + 24$

$= 32$

$y(2) = 32$

$y(3) = x(k)\, h(3-k)$

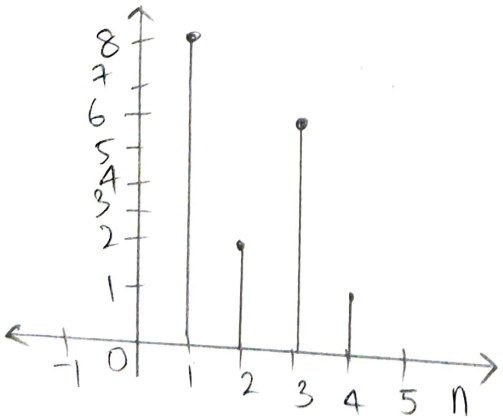


$= (8 \times 1) + (2 \times 4) + (6 \times 6) + 1$

$= 8 + 8 + 36 + 1$

$y(3) = 53$

$y(4) = x(k)\, h(4-k)$

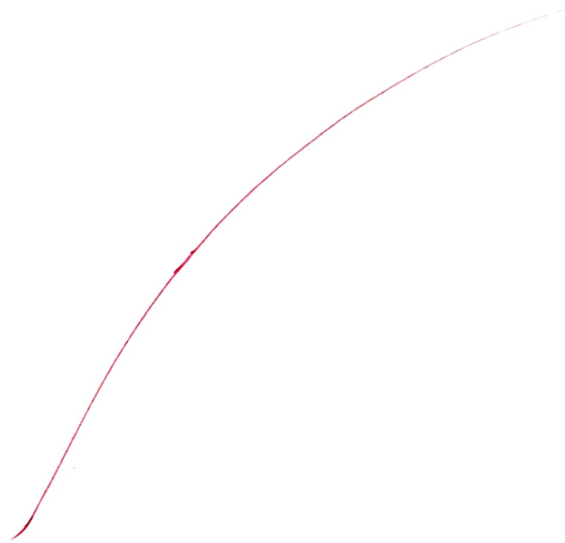


$= (4 \times 8) + (6 \times 2) + (6)$

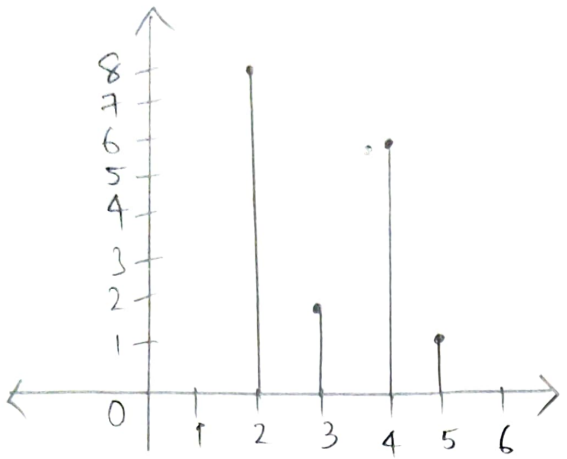
$= 32 + 12 + 6$

$y(4) = 50.$

```
disp (y, 'convolution sum using direct formula
    Method =')
subplot (3,1,3);
plot2d3 (y);
title('Graphical Representation of output signal y');
```

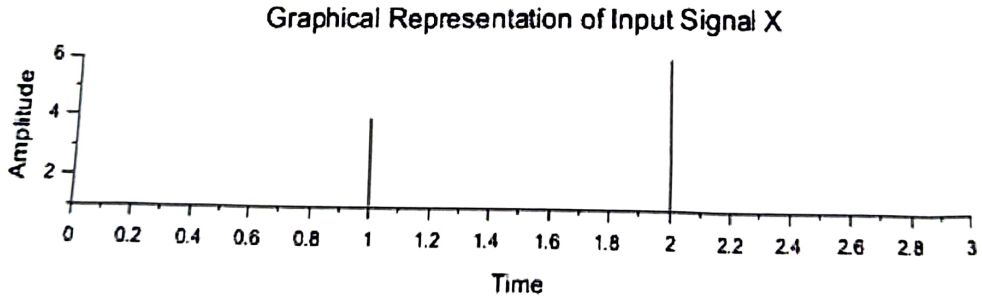


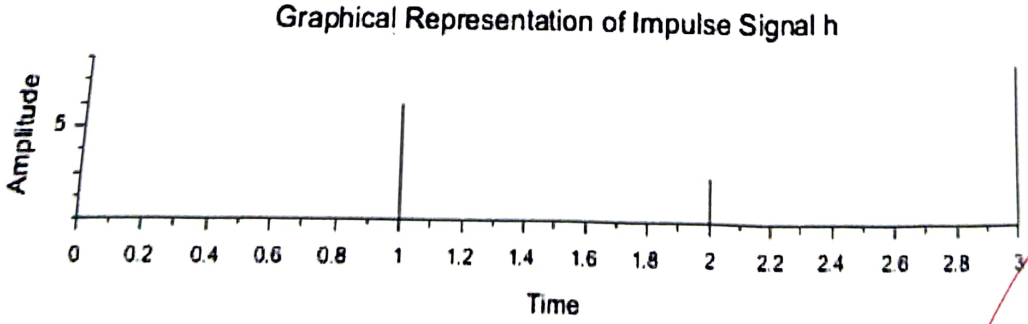
$y(5) = x(k)\,h(5-k)$

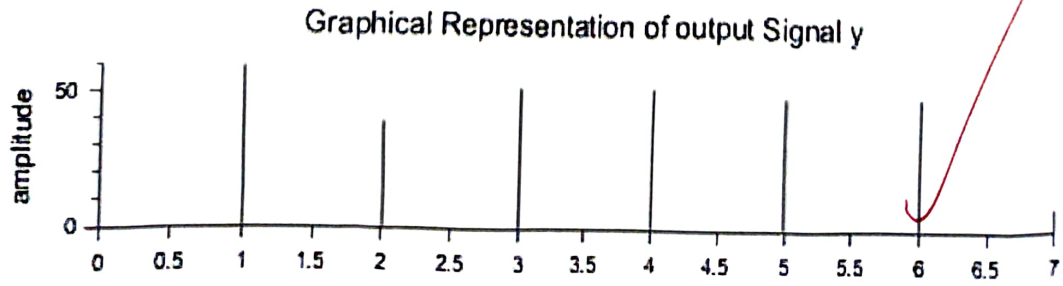


$= (6 \times 8) + (1 \times 2) \Rightarrow 48 + 2$

$y(5) = 50.$







RESULT:-

Thus, the linear convolution and circular convolution using DFT.